BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: MRD-X-21 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** MRD-X-21 (A) VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. If the universal set U = {x|x is a positive odd integer less then 30}, R = {1, 5, 7} and S = {1, 3, 7, 11, 13}, how many elements are in (R ∩ S)' ?

(A) 15
(B) 13
(C) 7
(D) 2

Q2. The nth root of product of 'n' of values is called .....

(A) Arithmetic mean
(B) Geometric mean
(C) Harmonic mean
(D) Standard deviation

Q3. If $x^2 - 1/4 = 0$, the solution is .....

(A) $x = \pm 1/2$
(B) $x = \pm 1/4$
(C) $x = \pm 1/8$
(D) $x = \pm 1/16$

Q4. One and only one ..... can pass through three non-collinear points.

(A) Rectangle
(B) Square
(C) Circle
(D) All of these

Q5. ..... tangents can be drawn from a point outside the circle.

(A) 4
(B) 3
(C) 2
(D) 1

Q6. The angle in a segment less than a semi-circle is ..... a right angle.

(A) Greater than
(B) Less than
(C) Equal to
(D) Less than or equal to

Q7. A chord 40 units long is contained in a circle of radius 25 cm. The distance of the chord from the centre of the circle is .....

(A) 15 units
(B) 31.2 units
(C) 47.1 units
(D) 50.1 units

Q8. $1 + \tan^2\theta =$ .....

(A) $\sin^2\theta$
(B) $\cos^2\theta$
(C) $\text{cosec}^2\theta$
(D) $\sec^2\theta$

Q9. In the given set of data 5, 7, 7, 5, 3, 7, 2, 8, 2 mode is .....

(A) 9
(B) 5
(C) 7
(D) 2

Q10. Product of the roots of the equation $3x^2 + 2x - 5 = 0$ is .....

(A) $-2/3$
(B) $-5/3$
(C) -15
(D) None of these

Q11. $(x^2 + 2)/(x^2 + 2x + 2)$ is ..... fraction.

(A) Proper
(B) Improper
(C) Irrational
(D) None of these

Q12. Apollonius is the name of a .....

(A) City
(B) Town
(C) Country
(D) Mathematician

Q13. If $x \propto 1/y$ and $y \propto 1/z$, then .....

(A) $y \propto 1/z$
(B) $x \propto z$
(C) $xy \propto z$
(D) $xz \propto y$

Q14. If $(x + 1)(x - 5) = 0$, then the solutions are .....

(A) $x = 1, -5$
(B) $x = 1, 5$
(C) $x = -1, -5$
(D) $x = -1, 5$

Q15. If an object is above the level of observer, then the angle formed between the horizontal line and observer's line of sight is called .....

(A) Angle of depression
(B) Angle of elevation
(C) Reflex angle
(D) None of these

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 36

Q2. Attempt any nine (9) of the following parts. All parts carry equal marks.

(i) Solve the equation by using quadratic formula: $x^2/3 - x/12 = 1/24$

(ii) For what value of k, the roots of the equation are equal: $x^2 + kx + 9 = 0$

(iii) If α, β are the roots of the equation $2x^2 + 3x + 1 = 0$, then find the value of $\alpha/\beta + \beta/\alpha$

(iv) The sides of a right triangle in cm are $(x - 1)$, x, $(x + 1)$. Find the sides of triangle.

(v) If y varies jointly as x and z and $y = 33$ when $x = 9$ and $z = 12$. Find y when $x = 16$ and $z = 22$.

(vi) If $(x-y)/z = (y-z)/x = (z-x)/y$, then prove that $x = y = z$ where x,y,z are non-zero numbers and $x + y + z \neq 0$

(vii) Resolve into partial fraction: $x/(x^2 + 4x - 5)$

(viii) If U = {1,2,3,.....,15}, A = {2,4,6,.....,14} and B = {1,3,5,.....,15}. Find A' ∩ B'

(ix) A set of data contains the values as 148, 145, 160, 157, 156 and 160. Show that Mode > Median > Mean.

(x) Prove that: $(\sec^2\theta - 1)\cos^2\theta = \sin^2\theta$

(xi) Convert the following radian measures of angles into the measure of degrees: (a) 2 radians (b) π/6 radians

(xii) If x = {1, 2, 3, 4} and y = {5, 6, 7, 8}, then write a function from x to y which is neither one-one nor onto.

Section-C

Maximum Marks: 24

> Note: Attempt any three (3) questions. All questions carry equal marks.

Q3. Prove that: One and only one circle can pass through three non-collinear points.

Q4. Prove that: If a line is drawn perpendicular to a radial segment of a circle at its outer end point, it is tangent to the circle at that point.

Q5. Prove that: If two angles subtended by two chords of congruent circles at the corresponding centres are equal in measures, then the chords are equal in measures.

Q6. Draw ∠ABC of measure 60°. Construct a circle having radius 2.5 cm and touching the arms of the angle.